

Engineering Design Study

Sequential Gearbox

Mechanical Design Report

Personal Engineering Project

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1 Introduction

1.1 Project Context

This project consists of the mechanical design and verification of a sequential gearbox intended for high-torque automotive applications, with a particular focus on off-road and rally-type operating conditions. The work focuses on the dimensioning and validation of the main mechanical components, including shafts, gears, torque transmission elements, and bearings, under realistic loading conditions. The gearbox is designed using standard engineering methods and established design references, with an emphasis on structural integrity, fatigue resistance, and mechanical reliability. Computer-aided design tools are used to define the geometry and assembly of the system, while analytical calculations are employed to assess stresses, safety factors, and component life. The objective of the project is to develop a gearbox design representative of real-world engineering practice.

1.2 Motivation

The idea behind this project is to apply mechanical engineering design methods to a complete drivetrain subsystem rather than to isolated components. Sequential gearboxes involve the interaction of shafts, gears, bearings, and torque transmission elements under combined loading and fatigue conditions, making them a relevant case study for power transmission design. They are known for their purely mechanical mechanism. This project aims to bridge the gap between academic theory and practical engineering by using recognized design standards, realistic load cases, and iterative dimensioning to develop a coherent and mechanically robust gearbox concept.

1.3 Design Objectives

The primary objectives of the sequential gearbox design are summarized as follows:

- Safely transmit the specified engine torque under high-load operating conditions, including shock loading representative of off-road and rally-type use.
- Ensure adequate fatigue life for all rotating and torque-transmitting components over the intended service life.
- Achieve reliable sequential gear engagement through mechanically robust engagement elements.
- Use realistic materials, heat treatments, and manufacturing constraints consistent with industrial practice.
- Apply recognized mechanical design standards and references for stress, fatigue, and life verification.
- Maintain a compact and mechanically coherent layout suitable for automotive integration.
- Focus on the development of a technically realistic design, rather than weight minimization, conceptual novelty, or performance optimization beyond standard engineering practice.

2 Technical Specifications

2.1 Engine Torque and Speed Range

The gearbox is designed to operate with an internal combustion engine delivering high torque at relatively moderate rotational speeds, representative of off-road and rally-type automotive applications. The design calculations are based on a nominal peak engine torque of

$$T_{\text{eng}} = 450 \text{ N}\cdot\text{m}$$

with an engine speed range extending up to a redline speed of

$$n_{\text{red}} = 6500 \text{ rpm}$$

These values are used as reference inputs for the dimensioning of shafts, gears, bearings, and torque transmission elements, ensuring adequate strength and fatigue resistance under the most demanding operating conditions.

2.2 Gear Set and Ratios

The selected gear pairs were obtained after multiple iterations to satisfy the targeted ratio spread and a constant center distance constraint for the forward gears. The final gear data are summarized in Table 1. A dedicated layout is used for the reverse gear train to satisfy the center distance requirement without collinearity of the input, idler, and output gears.

Table 1: Gear specifications and ratios

Gear	Module	z_{in}	z_{out}	Ratio i_{gear}	Center distance [mm]
1	3.0	25	63	2.52	132
2	3.0	31	57	1.8387	132
3	3.0	37	51	1.378	132
4	3.0	41	47	1.1463	132
5	3.0	49	39	0.7959	132
Reverse	2.5	31	68	2.1935	132
Idler	2.5	23	–	–	–

2.3 Assumptions for Speed and Wheel Torque

To evaluate vehicle speed and wheel torque across gears, the following assumptions are used:

- Final drive (differential) ratio: $i_{\text{final}} = 4.8$
- Differential efficiency: $\eta_{\text{final}} = 0.9$
- Tire size: 225/40 R18, resulting in an approximate tire diameter $D \approx 0.6372 \text{ m}$ and circumference $C_{\text{tire}} \approx 2.0 \text{ m}$

2.4 Speed and Torque Across Gears

Wheel speed and vehicle speed at engine speed n_{eng} are computed using:

$$n_{\text{wheel}} = \frac{n_{\text{eng}}}{i_{\text{gear}} i_{\text{final}}}, \quad v = \frac{n_{\text{wheel}} C_{\text{tire}} 60}{1000},$$

where v is expressed in km/h when n_{wheel} is in rpm and C_{tire} in meters. The gearbox output torque and wheel torque are computed as:

$$T_{\text{out}} = T_{\text{eng}} i_{\text{gear}}, \quad T_{\text{wheel}} = T_{\text{eng}} i_{\text{gear}} i_{\text{final}} \eta_{\text{final}}.$$

Using $T_{\text{eng}} = 450 \text{ N} \cdot \text{m}$ and $n_{\text{eng}} = n_{\text{red}} = 6500 \text{ rpm}$, the results are summarized in Table 2.

Table 2: Vehicle speed and wheel torque at redline ($n_{\text{eng}} = 6500 \text{ rpm}$).

Gear	Speed [km/h]	Wheel torque T_{wheel} [N·m]
1	64.54	4898.88
2	88.46	3574.43
3	118.03	2678.83
4	141.89	2228.41
5	204.36	1547.23

The values obtained are considered acceptable for the intended application. The design focus is not maximum top speed, but rather mid-range drivability, torque availability, and acceleration under severe operating conditions.

2.5 Reference Service Life and Operating Assumptions

For the purpose of the gearbox design and subsequent mechanical verification, a reference service life and representative operating conditions are defined. These assumptions are used consistently throughout the shaft, gear, and bearing calculations.

The target service life of the gearbox is set to

$$L_{\text{km}} = 40\,000 \text{ km}.$$

Assuming an average vehicle speed of

$$v_{\text{avg}} = 60 \text{ km/h},$$

the corresponding operating life in hours is

$$L_h = \frac{40\,000}{60} \approx 667 \text{ h}.$$

A mixed usage profile is assumed, combining aggressive off-road driving, rally-type operation, and normal cruising. The distribution of operating time across the forward gears is represented by the time fractions listed in Table 3. These values are used later to define representative loading and duty cycles.

Table 3: Assumed time fraction per forward gear.

Gear	Time fraction
Gear 1	0.10
Gear 2	0.20
Gear 3	0.20
Gear 4	0.25
Gear 5	0.25

2.6 Materials Summary

The materials selected for the main gearbox components are summarized in Table 4. Detailed justification and verification are provided in the corresponding design sections.

Table 4: Summary of materials used for main gearbox components.

Component	Material
Shafts	AISI 4140 (Q&T at 425 °C)
Gears	18CrNiMo7-6 (case-hardened)
Dog Rings	18CrNiMo7-6
Keys	AISI 4340 (Normalized)

3 Gearbox Architecture and Layout

This section describes the overall architecture of the sequential gearbox and the main constraints that governed its layout. The objective is to present the structural organization of the transmission and the interaction between its main components prior to detailed mechanical verification.

3.1 Sequential Gearbox Concept

The gearbox follows a sequential shifting concept, where gears are engaged in a fixed order through axial movement of engagement elements called "dog rings". Gear selection is performed step-by-step, without the possibility of directly skipping gears. This architecture is well suited for applications requiring rapid and robust gear changes under high load conditions, such as off-road and rally-type use.

3.2 Shaft Arrangement

The gearbox architecture is based on an input shaft and an output shaft, with gear pairs arranged to achieve the required transmission ratios. Forward gears are laid out to maintain a constant center distance, simplifying shaft geometry and bearing arrangement. The output gears are designed to rotate freely on the output shaft and are selectively engaged through dedicated engagement elements, allowing individual gear selection without permanent coupling.

3.3 Gear Ratio Selection Constraints

The selection of the gear ratios was driven by multiple constraints, including the desired ratio spread, acceptable vehicle speeds, torque availability across gears, and geometric limitations imposed by a fixed center distance. Additional constraints included the use of standard gear modules, realistic tooth counts, and compatibility with the chosen shaft layout, as well as ensuring even wear across all gear teeth. Several candidate gear sets were evaluated, and the final configuration was retained based on its ability to satisfy these constraints while maintaining a mechanically coherent architecture.

3.3.1 Gear Ratio and Center Distance Logic

The transmission ratio of each gear pair is defined by the ratio of the number of teeth of the output gear to that of the input gear,

$$i_{\text{gear}} = \frac{z_{\text{out}}}{z_{\text{in}}}.$$

For a given gear module m , the pitch diameter of a gear is directly proportional to its number of teeth,

$$d = m z.$$

The center distance between the input and output shafts is equal to the sum of the pitch radii of the mating gears and can be expressed as

$$a = \frac{d_{\text{in}} + d_{\text{out}}}{2} = \frac{m}{2} (z_{\text{in}} + z_{\text{out}}).$$

By expressing the number of teeth of the output gear as a function of the transmission ratio,

$$z_{\text{out}} = i_{\text{gear}} z_{\text{in}},$$

the center distance relation becomes

$$a = \frac{m}{2} z_{\text{in}} (1 + i_{\text{gear}}).$$

The gear set selection was initiated with the first gear, which was sized to achieve the desired gearbox output torque range of approximately 1300–1500 N · m to obtain the desired torque at wheel value, while maintaining realistic tooth counts and standard gear modules. The center distance was fixed based on the first gear.

Using the obtained center distance relation, the gear selection procedure consists of choosing a target transmission ratio and determining an admissible number of teeth for the input gear. The corresponding number of teeth of the output gear is then calculated from the ratio relation.

The resulting tooth numbers are checked to ensure integer values, realistic gear proportions, and favorable meshing conditions, in particular by avoiding common divisors between the mating gears in order to promote uniform tooth wear.

Once a candidate pair of tooth numbers is obtained, the center distance and the resulting transmission ratio are recomputed to verify consistency. If the geometric and kinematic conditions are satisfied, the gear pair is retained and the procedure is repeated for the next gear.

3.4 Reverse Gear Arrangement

Unlike the forward gears, the reverse gear train requires an additional idler gear to invert the direction of rotation. To satisfy the center distance constraint while avoiding co-linearity between the input, output, and idler gears, a dedicated reverse gear layout is implemented. This arrangement allows the reverse gear to meet geometric requirements.

The resulting arrangement is the following:

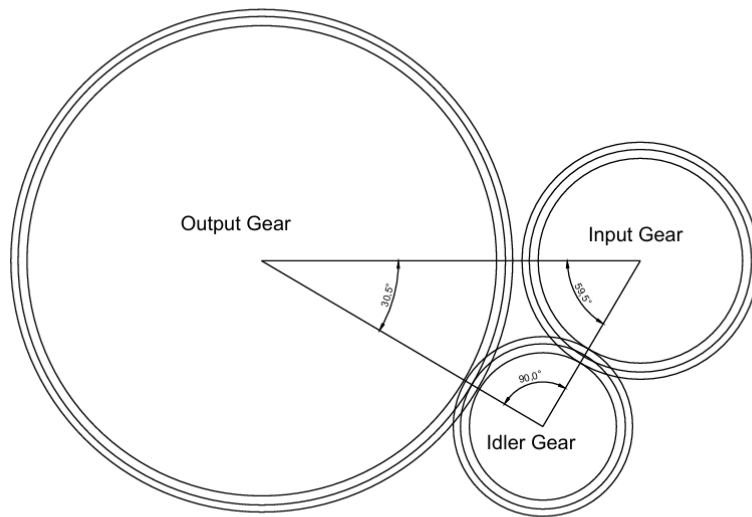


Figure 1: Reverse gear arrangement with offset idler gear to satisfy the center distance constraint.

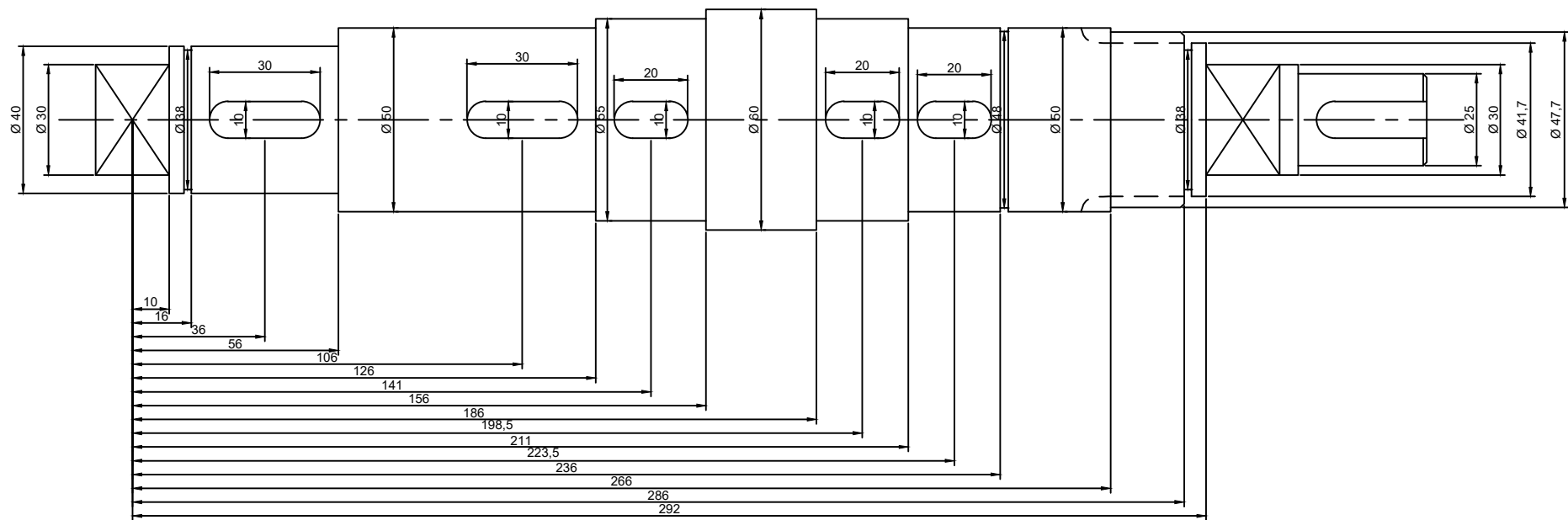
The relative positioning of the gears is defined by angle values of 59.5° and 30.5° .

4 Shaft Design and Verification

4.1 Shaft Layout and Geometry Definition

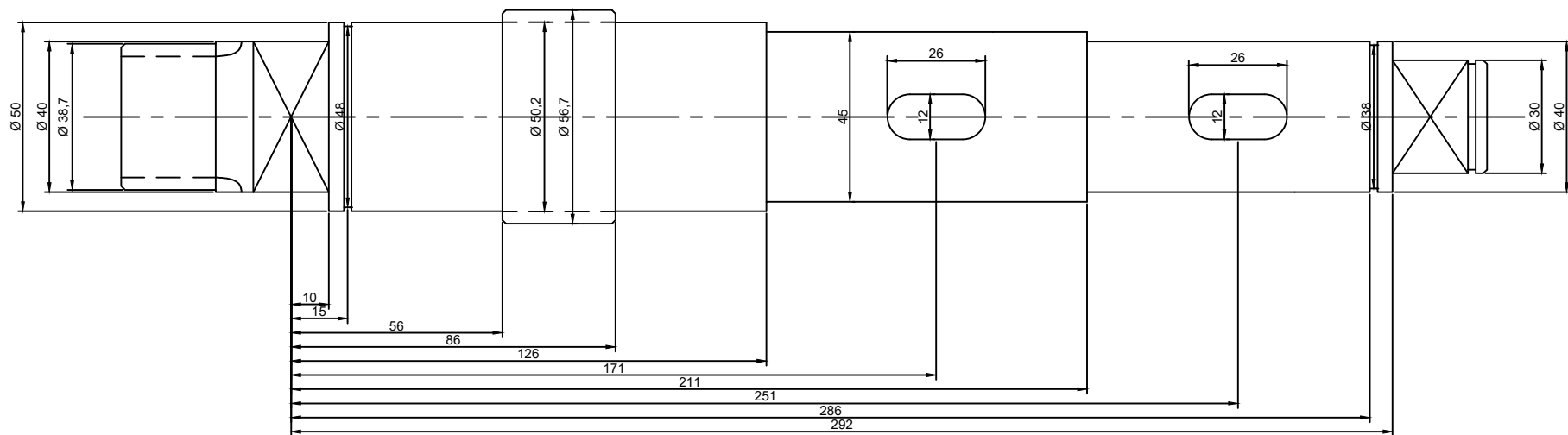
The shaft geometry is defined prior to stress and fatigue verification in order to establish the locations of gears, bearings, and torque transmission elements. This preliminary layout is used to identify critical sections, define load application points, and provide a consistent geometric basis for the analytical calculations presented in the following sections. The geometry presented at this stage corresponds to an initial design intended for verification purposes and does not represent the final manufacturing drawing.

The dimensions in this drawing are used for moment diagrams and all fatigue analysis. The dimensions were chosen based on previous failed designs and they were selected in a way to avoid decimal numbers while always maintaining conservative results.



Input Shaft

Scale 1:1



Output Shaft

Scale 1:1

4.2 Load Cases on Shafts

Due to the sequential nature of the gearbox, only one gear pair is engaged and transmitting torque at any given time. Although all gears are in continuous rotation, torque is transferred exclusively through the engaged gear, while the remaining gears rotate freely without carrying load. Consequently, each gear engagement is treated as an independent load case.

The shaft loading analysis is therefore performed by considering each forward gear and the reverse gear separately. For each load case, the transmitted torque is assumed to be constant along the shaft segment between the input and the engaged gear, while bending loads arise from the radial forces generated at the gear mesh and transmitted to the shaft through the bearings.

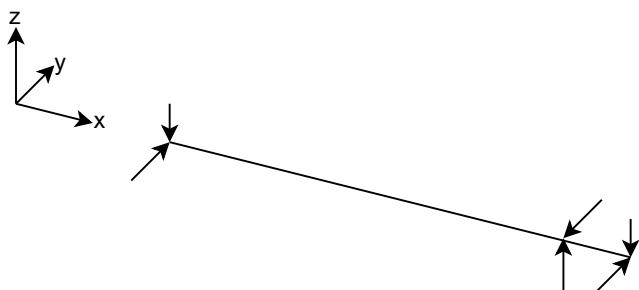
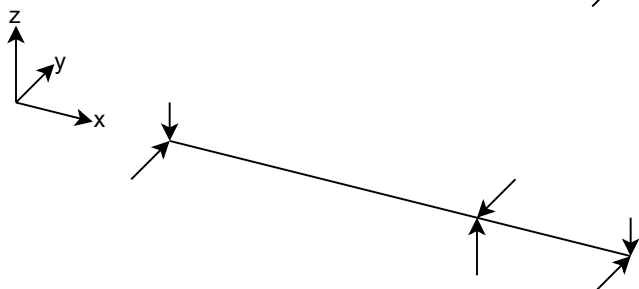
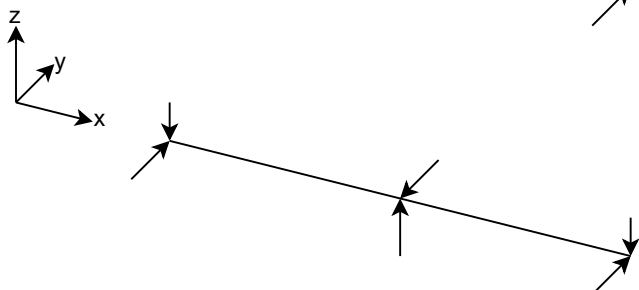
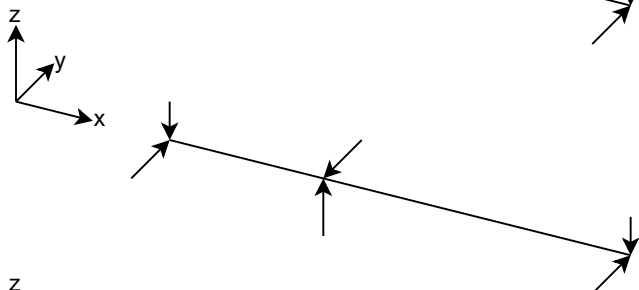
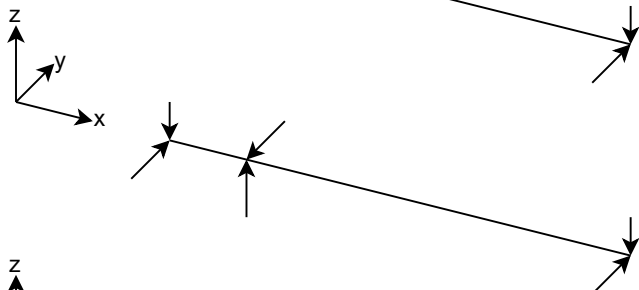
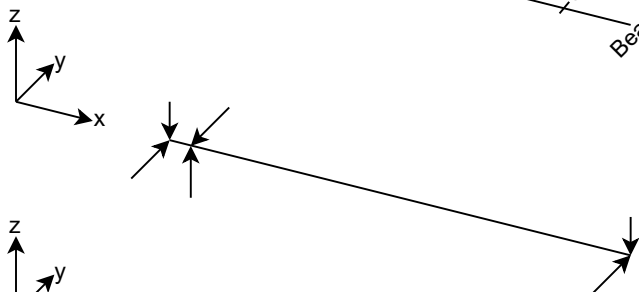
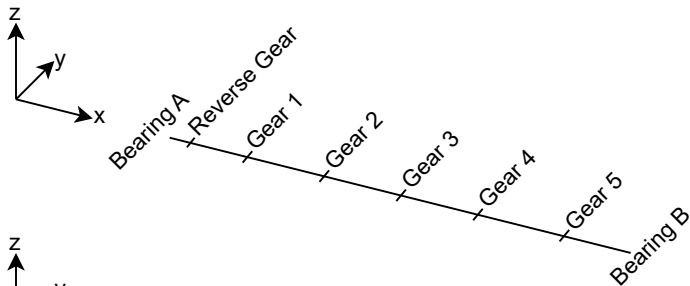
For a given gear engagement, the applied loads are defined by:

- The transmitted torque corresponding to the engaged gear ratio.
- The radial gear mesh forces acting at the gear location.
- The reaction forces at the bearing supports.

Free-body diagrams are constructed for each shaft configuration, and the resulting bending moment distributions are obtained. From these, bending moment diagrams are generated in the principal planes, allowing me to obtain the absolute maximum moment at each critical point on the shaft (grooves, shoulder, keyseats, ...) required for shaft fatigue analysis. This procedure is repeated for each gear engagement.

For the output shaft, the same bending moment diagrams obtained for the input shaft are used as a basis for the stress and fatigue analysis. This is justified by the fact that, for each engaged gear, the radial gear mesh forces acting on the two mating gears are equal in magnitude and opposite in direction.

In addition, the axial position of gears and bearings are the same on the input and output shaft. Consequently, the resulting bending loads transmitted to the input and output shafts are identical in magnitude for a given gear engagement.



Since the analysis focuses on the magnitude of bending moments and the resulting stress envelopes, the same moment diagrams can be conservatively applied to both shafts. So in practice, the moment diagrams on the output shaft should be in the opposite direction as those on the input. But since I will be using the absolute maximum moment value, the sign of the moment has no effect on results. Since the gear engagement mechanism deals with 2 gears at a time, I used the moment diagrams for the more critical case between each pair of gears.

4.3 Bending Moments and Torsional Loading at Critical Sections

For subsequent stress and fatigue verification, the bending moment and torque values are summarized at a set of predefined critical shaft features (shoulders, bearing seats, grooves, keyseats, and spline locations). After many attempts at different designs, the most critical case that was controlling the design was when gear 1 is engaged. So for this reason, the values that will be tabulated and shown in this report are going to be for this case, while always providing full results for other cases in the additional documents to this report.

4.3.1 Input Shaft

For each feature, the absolute maximum bending moment is calculated, which corresponds to the resultant bending moment magnitude, $M = \sqrt{M_x^2 + M_y^2}$.

Table 5: Bending moments and torsional loading at critical shaft features – Gear 1 engaged – input shaft

Feature	Location [mm]	M_{xy} [N·m]	M_{xz} [N·m]	$M_{\max}$ [N·m]	$T_{\max}$ [N·m]
Shoulder bearing A	10	26.64	-73.18	77.88	450
Groove	15	39.96	-109.78	116.82	450
Keyseat (reverse gear)	36	95.89	-263.46	280.37	450
Shoulder after reverse	56	149.17	-409.83	436.13	450
Keyseat gear 1	106	282.35	-775.75	825.54	450
Shoulder after gear 1	126	253.54	-696.59	741.30	450
Keyseat gear 2	141	231.93	-637.22	678.12	450
Shoulder after gear 2	156	210.32	-577.85	614.94	450
Shoulder before gear 3	186	167.11	-459.12	488.58	450
Keyseat gear 3	198	149.82	-411.62	438.04	450
Shoulder after gear 3	211	131.09	-360.17	383.28	450
Keyseat gear 4	223	113.80	-312.67	332.74	450
Groove	237	93.64	-257.26	273.77	450
Shoulder before gear 5	266	51.86	-142.48	151.63	450
Groove	286	23.05	-63.33	67.39	450
Shoulder bearing B	292	14.41	-39.58	42.12	450

4.3.2 Output Shaft

Table 6: Bending moments and torsional loading at critical shaft features – Gear 1 engaged – output shaft

Feature	Location [mm]	M_{xy} [Nm]	M_{xz} [Nm]	$M_{\max}$ [Nm]	$T_{\max}$ [Nm]
Shoulder bearing A	10.00	26.64	-73.18	77.88	450.00
Groove	15.00	39.96	-109.78	116.82	450.00
Shoulder after reverse	56.00	149.17	-409.83	436.13	450.00
Shoulder before gear 1	86.00	229.08	-629.38	669.77	450.00
Shoulder after gear 1	126.00	253.54	-696.59	741.30	450.00
Dog sleeve keyseat gear 2–3	171.00	188.71	-518.49	551.76	450.00
Shoulder after gear 3	211.00	131.09	-360.17	383.28	450.00
Dog sleeve keyseat gear 4–5	251.00	73.47	-201.85	214.81	450.00
Groove	286.00	23.05	-63.33	67.39	450.00
Shoulder bearing B	292.00	14.41	-39.58	42.12	450.00

4.4 Shaft Fatigue Analysis

4.4.1 Shaft Material

The shafts are assumed to be manufactured from AISI 4140 steel, quenched and tempered at 425 °C. The mechanical properties used for the stress and fatigue calculations are summarized below:

- **Material:** AISI 4140 steel
- **Heat treatment:** quenched and tempered at 425 °C
- **Ultimate tensile strength:** $S_{ut} = 1250$ MPa
- **Yield strength:** $S_y = 1140$ MPa

4.4.2 Alternating and Mid-range Parameters

Before applying fatigue failure criteria, the nature of the bending and torsional loading acting on the gearbox shafts is defined. In the following, the subscripts a and m denote the alternating and midrange components of the applied loads, respectively.

Bending Loading

The bending moments acting on the gearbox shafts are generated by the radial forces arising from gear mesh interactions. These forces are fixed in space, as they are transmitted through the gearbox housing and bearing supports, while the shafts rotate relative to them.

As a consequence, a material point on the shaft experiences a reversal of bending stress every half revolution. From a fatigue perspective, the shaft therefore behaves as a rotating beam subjected to fully reversed bending. Accordingly, the bending moment components are defined as:

$$M_a = M_{\max}, \quad M_m = 0,$$

where $M_{\max}$ is the resultant bending moment obtained from the bending moment diagrams.

This assumption applies to both the input and output shafts and corresponds to the classical rotating-shaft fatigue model described in standard mechanical design references.

Torsional Loading

Torque is transmitted through gears that are rotationally fixed to the shafts. For the input shaft, torque is supplied by the engine and is always transmitted in the same direction. Although torque fluctuations may occur due to engine firing pulses, the transmitted torque does not reverse sign.

For the output shaft, torque direction may change during acceleration, engine braking, or overrun conditions. However, in the absence of detailed vehicle dynamics and braking torque data, a simplified and consistent assumption is adopted for the present design study.

For both the input and output shafts, the torsional loading is therefore treated as steady torsion, characterized by:

$$T_m = T_{\max}, \quad T_a = 0,$$

where $T_{\max}$ is the maximum transmitted torque for the considered gear engagement.

This treatment is commonly adopted in gearbox shaft fatigue analyses and is consistent with standard mechanical design practice.

Summary of Fatigue Loading Model

Based on the above considerations, the fatigue analysis of the gearbox shafts is performed using the following loading model:

- **Bending (fully reversed):**

$$M_a = M_{\max}, \quad M_m = 0$$

- **Torsion (steady):**

$$T_m = T_{\max}, \quad T_a = 0$$

This loading model represents a realistic and conservative approximation of the operating conditions of a sequential gearbox and provides a consistent basis for the application of fatigue failure criteria in the following sections.

4.4.3 Fatigue Failure Criteria

Several fatigue failure criteria are available for the evaluation of components subjected to combined mid-range and alternating stresses. These criteria differ in their treatment of material behavior in the finite life and endurance regimes and therefore lead to different safety factor predictions. In the present work, multiple criteria are evaluated in order to provide a comprehensive comparison of the fatigue response.

Safety factors are calculated using the Goodman, Gerber, Soderberg, and ASME elliptic criteria. While results from all criteria are reported for completeness, the Goodman criterion is retained as the governing design criterion. This choice is made because of its conservative nature and its widespread use in practical mechanical design, particularly for steel components subjected to combined bending and torsional loading.

Goodman Criterion

Safety Factor Form:

$$\frac{1}{n} = \frac{16}{\pi d^3} \left[\frac{1}{S_e} \left(4 (K_f M_a)^2 + 3 (K_{fs} T_a)^2 \right)^{1/2} + \frac{1}{S_{ut}} \left(4 (K_f M_m)^2 + 3 (K_{fs} T_m)^2 \right)^{1/2} \right] \quad (1)$$

Diameter form:

$$d = \left[\frac{16n}{\pi} \left(\frac{1}{S_e} \left(4 (K_f M_a)^2 + 3 (K_{fs} T_a)^2 \right)^{1/2} + \frac{1}{S_{ut}} \left(4 (K_f M_m)^2 + 3 (K_{fs} T_m)^2 \right)^{1/2} \right) \right]^{1/3} \quad (2)$$

where:

- n is the fatigue safety factor
- d is the shaft diameter
- M_a, M_m are the alternating and midrange bending moments
- T_a, T_m are the alternating and midrange torques
- K_f is the fatigue stress concentration factor in bending
- K_{fs} is the fatigue stress concentration factor in torsion
- S_e is the corrected endurance limit
- S_{ut} is the ultimate tensile strength

Gerber Criterion

Safety Factor Form:

$$\frac{1}{n} = \frac{8A}{\pi d^3 S_e} \left[1 + \sqrt{1 + \left(\frac{2BS_e}{AS_{ut}} \right)^2} \right] \quad (3)$$

Diameter Form:

$$d = \left[\frac{8nA}{\pi S_e} \left(1 + \sqrt{1 + \left(\frac{2BS_e}{AS_{ut}} \right)^2} \right) \right]^{1/3} \quad (4)$$

where

$$A = \sqrt{4 (K_f M_a)^2 + 3 (K_{fs} T_a)^2} \quad (5)$$

$$B = \sqrt{4 (K_f M_m)^2 + 3 (K_{fs} T_m)^2} \quad (6)$$

ASME Elliptic Criterion

Safety Factor Form:

$$\frac{1}{n} = \frac{16}{\pi d^3} \left[4 \left(\frac{K_f M_a}{S_e} \right)^2 + 3 \left(\frac{K_{fs} T_a}{S_e} \right)^2 + 4 \left(\frac{K_f M_m}{S_y} \right)^2 + 3 \left(\frac{K_{fs} T_m}{S_y} \right)^2 \right]^{1/2} \quad (7)$$

Diameter Form:

$$d = \left[\frac{16n}{\pi} \left(4 \left(\frac{K_f M_a}{S_e} \right)^2 + 3 \left(\frac{K_{fs} T_a}{S_e} \right)^2 + 4 \left(\frac{K_f M_m}{S_y} \right)^2 + 3 \left(\frac{K_{fs} T_m}{S_y} \right)^2 \right)^{1/2} \right]^{1/3} \quad (8)$$

Soderberg Criterion

Safety Factor Form:

$$\frac{1}{n} = \frac{16}{\pi d^3} \left[\frac{1}{S_e} \left(4 (K_f M_a)^2 + 3 (K_{fs} T_a)^2 \right)^{1/2} + \frac{1}{S_y} \left(4 (K_f M_m)^2 + 3 (K_{fs} T_m)^2 \right)^{1/2} \right] \quad (9)$$

Diameter Form:

$$d = \left[\frac{16n}{\pi} \left(\frac{1}{S_e} \left(4 (K_f M_a)^2 + 3 (K_{fs} T_a)^2 \right)^{1/2} + \frac{1}{S_y} \left(4 (K_f M_m)^2 + 3 (K_{fs} T_m)^2 \right)^{1/2} \right) \right]^{1/3} \quad (10)$$

Corrected Endurance Limit S_e

The endurance limit of the shaft material is evaluated using the modified endurance limit expression

$$S_e = k_a k_b k_c k_d k_e k_f S'_e$$

where:

- k_a = surface condition modification factor, *Shigley's Mechanical Engineering Design*, Table 6-2
- k_b = size modification factor, *Shigley's Mechanical Engineering Design*, Equation 6-20
- k_c = load modification factor, *Shigley's Mechanical Engineering Design*, Table 6-4
- k_d = temperature modification factor, *Shigley's Mechanical Engineering Design*, Equation 6-27
- k_e = reliability factor, *Shigley's Mechanical Engineering Design*, Table 6-5
- k_f = miscellaneous-effects modification factor
- S'_e = rotary-beam test specimen endurance limit

The unmodified endurance limit for steel is defined as

$$S'_e = \begin{cases} 0.5 S_{ut} & S_{ut} \leq 1400 \text{ MPa} \\ 700 \text{ MPa} & S_{ut} > 1400 \text{ MPa} \end{cases}$$

For the material I chose

$$S_{ut} = 1250 \text{ MPa}$$

Thus

$$S'_e = 0.5 S_{ut}$$

Fatigue Stress Concentration Factors

Geometric discontinuities such as shoulders, grooves, and keyways introduce stress concentrations that reduce fatigue strength. The fatigue stress concentration factors in bending and torsion, denoted by K_f and K_{fs} , respectively, are evaluated using the relations

$$K_f = 1 + q(K_t - 1), \quad K_{fs} = 1 + q_s(K_{ts} - 1)$$

where K_t and K_{ts} are the theoretical stress concentration factors, and q and q_s are the notch sensitivity factors for bending and torsion.

Corresponding values for K_t and K_{ts} are provided in *Shigley's Mechanical Engineering Design*, Figure A-15-8 and Figure A-15-9.

While q and q_s values are provided in *Shigley's Mechanical Engineering Design*, Figure 6-20 and Figure 6-21.

The obtained values for my shaft design are tabulated below.

Table 7: Theoretical stress concentration factors for the input shaft

Feature	Shaft diameter [mm]	Notch radius [mm]	K_t	K_{ts}
Shoulder bearing A	30.0	1.00	2.30	1.90
Groove	38.0	0.15	4.60	2.75
Keyseat (reverse gear)	40.0	0.90	2.14	3.00
Shoulder after reverse	40.0	1.50	1.90	1.80
Keyseat gear 1	50.0	0.90	2.14	3.00
Shoulder after gear 1	50.0	1.50	2.10	1.90
Keyseat gear 2	55.0	0.90	2.14	3.00
Shoulder after gear 2	55.0	1.50	2.18	1.45
Shoulder before gear 3	55.0	1.50	2.18	1.45
Keyseat gear 3	55.0	0.90	2.14	3.00
Shoulder after gear 3	50.0	1.50	2.10	1.90
Keyseat gear 4	50.0	0.90	2.14	3.00
Groove	48.0	0.15	4.50	2.50
Shoulder before gear 5	41.7	1.00	2.30	1.90
Groove	38.0	0.15	5.75	3.45
Shoulder bearing B	30.0	1.00	2.30	1.90

Table 8: Theoretical stress concentration factors for the output shaft

Feature	Shaft diameter [mm]	Notch radius [mm]	K_t	K_{ts}
Shoulder bearing A	40.0	1.00	2.40	2.00
Groove	48.0	0.15	4.50	2.50
Shoulder after reverse	50.0	1.00	2.30	1.65
Shoulder before gear 1	50.0	1.00	2.30	1.65
Shoulder after gear 1	45.0	1.00	2.25	1.65
Dog sleeve keyseat (gears 2–3)	45.0	0.90	2.14	3.00
Shoulder after gear 3	40.0	1.00	2.25	1.55
Dog sleeve keyseat (gears 4–5)	40.0	0.90	2.14	3.00
Groove	38.0	0.15	4.60	2.75
Shoulder bearing B	30.0	1.00	2.30	1.90

4.4.4 Input Shaft Safety Factors

The resulting fatigue failure safety factors for input shaft are tabulated below. A complete fatigue analysis was performed for all forward gears and reverse, treating each engagement as an independent load case. Among these, first gear produces the highest transmitted torque and bending moments on both the input and output shafts, resulting in the lowest fatigue safety factors. Consequently, the first gear represents the governing design case. For clarity and conciseness, detailed safety factor tables are presented only for the input and output shafts with first gear engaged. The results for the remaining gears exhibit higher safety margins and are therefore omitted from the main body of this report. However, all tables are available in the accompanying files.

Table 9: Fatigue safety factors for the **input shaft – Gear 1 engaged**

Feature	n_{Goodman}	n_{Gerber}	n_{ASME}	$n_{\text{Soderberg}}$
Shoulder bearing A	2.691	3.259	2.240	2.562
Groove (15 mm)	2.901	3.626	2.941	2.803
Keyseat reverse gear	2.706	3.381	2.806	2.620
Shoulder after reverse	2.438	2.895	2.739	2.394
Keyseat gear 1	2.344	2.710	2.621	2.311
Shoulder after gear 1	2.747	3.085	3.029	2.717
Keyseat gear 2	3.626	4.266	4.072	3.565
Shoulder after gear 2	4.258	4.733	4.664	4.216
Shoulder before gear 3	5.204	5.902	5.769	5.141
Keyseat gear 3	5.072	6.198	5.629	4.954
Shoulder after gear 3	4.746	5.668	5.330	4.657
Keyseat gear 4	4.661	5.791	5.005	4.529
Groove (237 mm)	3.292	3.903	3.699	3.234
Shoulder before gear 5	5.048	6.310	5.117	4.877
Groove (286 mm)	3.149	3.857	2.745	3.008
Shoulder bearing B	3.342	3.834	2.346	3.146

4.4.5 Output Shaft Safety Factors

For the output shaft, the first-gear load case is governed by the dog ring location, which is positioned between the first-gear and reverse-gear regions. Therefore, the critical sections reported in Table 10 correspond to the shaft span between Gear 1 and Reverse, where the transmitted torque is effectively carried.

Table 10: Fatigue safety factors for the **output shaft – Gear 1 load case.**

Feature	n_{Goodman}	n_{Gerber}	n_{ASME}	$n_{\text{Soderberg}}$
Shoulder bearing A	3.223	3.607	2.121	3.021
Groove (15 mm)	3.823	4.580	3.062	3.631
Shoulder after reverse	3.296	4.098	3.533	3.203
Shoulder before gear 1	2.431	2.942	2.719	2.380
Shoulder after gear 1	1.682	2.023	1.885	1.648
Dog sleeve keyseat (gears 2–3)	1.758	2.194	1.726	1.694
Shoulder after gear 3	1.929	2.407	2.027	1.870
Dog sleeve keyseat (gears 4–5)	1.895	2.219	1.412	1.791
Groove (286 mm)	2.136	2.417	1.445	2.006
Shoulder bearing B	1.601	1.714	0.947	1.489

^a Note: While the ASME Elliptic criterion yields a value of 0.947 due to its high sensitivity to low bending moments, the more conservative Goodman criteria confirm the design is safe.

5 Gears Analysis

5.1 Gear Materials

All gears in the gearbox are manufactured from a low-carbon alloy steel specifically developed for carburizing applications. The selected material is **18CrNiMo7-6**, widely used in high-performance automotive and industrial gear transmissions due to its excellent fatigue and contact strength after heat treatment.

Material selection

The material was selected to meet the following mechanical requirements:

- High contact fatigue strength (pitting resistance)
- High bending fatigue strength at the tooth root
- Tough and ductile core to withstand shock loads
- Excellent response to carburizing heat treatment

Heat treatment

The gears are carburized, quenched, and tempered, resulting in a hard wear-resistant surface layer and a tough core. Typical properties after heat treatment are:

- Surface hardness: 58–62 HRC
- Effective case depth: 0.8–1.2 mm
- Core hardness: 30–40 HRC

Mechanical strength properties

Representative strength values for carburized 18CrNiMo7-6 used in gear design are:

- Ultimate tensile strength (core): $S_{ut} \approx 1100\text{--}1300$ MPa
- Yield strength (core): $S_y \approx 850\text{--}1000$ MPa
- Bending fatigue limit: $S_e \approx 550\text{--}650$ MPa
- Contact fatigue limit (pitting resistance): $S_t \approx 1500\text{--}1700$ MPa
- Young's modulus: $E \approx 210$ GPa
- Poisson's ratio: $\nu \approx 0.3$

Additional useful properties

The material used is a grade 2 steel, with a quality number of 10.

Design implications

The high surface hardness and contact fatigue strength ensure safe operation under severe contact stresses, while the tough core provides resistance to shock loading and torque reversals typical of a sequential gearbox.

Even tho torque reversals were not treated in this project but the idea is to resist to this phenomena.

These strength values are fully compatible with AGMA-based bending and contact stress calculations used throughout this project.

5.2 Failure Safety Factors

This section presents the analytical verification of the gearbox gears with respect to strength and fatigue requirements.

Gear durability is governed by two primary and independent failure modes: *tooth root bending failure* and *surface contact failure*. Both mechanisms are critical in high-torque, high-cycle applications such as sequential gearboxes and must be evaluated separately.

Tooth root bending failure is associated with cyclic tensile stresses at the fillet region of the gear tooth and can lead to fatigue cracking and eventual tooth breakage. Surface contact failure, on the other hand, results from excessive contact stresses between mating gear teeth and manifests primarily as pitting or micropitting on the tooth flanks.

In this section, the safety factors against bending fatigue and contact fatigue are evaluated using established AGMA-based analytical formulations.

The safety factor I am after is between 1.1 and 1.3. The reason that lets me go that low on safety factor is actually because of the nature of the gearbox. The gearbox itself operates in harsh conditions, but it is not built to live as long as regular gearboxes. As mentioned before, the target service life of the gearbox is 40 000 km. Such safety factors levels are widely common in motorsport and high-performance gearbox design practice.

5.2.1 Failure by bending

The bending stress at the gear tooth root is given by:

$$\sigma_F = \frac{W_t K_o K_v K_s K_H K_B}{b m_n Y_J} \quad (11)$$

where:

- W_t : transmitted tangential load [N]
- K_o : overload factor
- K_v : dynamic factor
- K_s : size factor
- b : face width [mm]
- m_n : normal module [mm]
- K_H : load distribution factor
- K_B : rim thickness factor
- Y_J : geometry factor for bending strength (including root fillet effects)

The allowable bending stress is given by:

$$\sigma_{F,all} = \frac{S_t Y_N}{S_F Y_\theta Y_Z} \quad (12)$$

where:

- S_t : allowable bending stress number of the material [MPa]
- Y_N : stress-cycle factor for bending fatigue
- Y_θ : temperature factor
- Y_Z : reliability factor
- S_F : bending safety factor (AGMA factor of safety)

Resulting bending safety factor:

$$S_F = \frac{\sigma_{F,all}}{\sigma_F} \quad (13)$$

5.2.2 Failure by wear (contact)

The contact stress between mating gear teeth is calculated as:

$$\sigma_H = Z_E \sqrt{\frac{W_t K_o K_v K_s K_H Z_R}{b d_p Z_I}} \quad (14)$$

where:

- Z_E : elastic coefficient [$\sqrt{N/mm^2}$]
- d_p : pitch diameter of the pinion [mm]
- Z_R : surface condition factor
- Z_I : geometry factor for pitting resistance

The allowable contact stress is defined as:

$$\sigma_{H,all} = \frac{S_c Z_N Z_W}{S_H Y_\theta Y_Z} \quad (15)$$

where:

- S_c : allowable contact stress number of the material [MPa]
- Z_N : stress-cycle factor for contact fatigue
- Z_W : hardness ratio factor for pitting resistance
- Y_θ : temperature factor
- Y_Z : reliability factor
- S_H : contact safety factor (AGMA factor of safety)

The resulting contact safety factor:

$$S_H = \frac{\sigma_{H,all}}{\sigma_H} \quad (16)$$

5.2.3 Input Parameters and Assumptions

Overload Factor K_o

The power source is considered uniform (constant torque all the time), with moderate shock in terms of operating conditions, so the resulting overload factor value is $K_o = 1.25$.

Information regarding this parameter are provided in *Shigley's Mechanical Engineering Design*, Figure 14–17.

Dynamic Factor K_v

The dynamic factor K_v accounts for the increase in dynamic loads caused by gear inaccuracies, manufacturing tolerances, and vibration effects. In AGMA methodology, K_v is related to the gear quality number Q_v and the pitch line velocity.

For metric units, the dynamic factor is given by:

$$K_v = \left(\frac{A + \sqrt{200V}}{A} \right)^B \quad (17)$$

where:

- V : pitch line velocity [m/s]
- Q_v : AGMA gear quality number

The coefficients A and B are defined as:

$$A = 50 + 56(1 - B) \quad (18)$$

$$B = 0.25 (12 - Q_v)^{2/3} \quad (19)$$

The pitch line velocity V is calculated as:

$$V = \frac{\pi d n}{60} \quad (20)$$

where:

- d : pitch diameter of the gear [m]
- n : rotational speed [rpm]

Higher values of the quality number Q_v correspond to more precise gears and result in lower dynamic factors.

Size factor k_s

AGMA does not provide definitive size factor values for all gear configurations. In the absence of a clearly detrimental size effect, AGMA recommends using a conservative value greater than or equal to unity. Accordingly, $K_s = 1$ may be assumed when size effects are negligible.

When size effects are considered, the size factor may be estimated using the AGMA-based empirical relation:

$$K_s = \frac{1}{k_b} = 1.192 \left(\frac{F\sqrt{Y}}{P} \right)^{0.0535} \quad (21)$$

where:

- F : face width [in]
- Y : Lewis form factor, given in *Shigley's Mechanical Engineering Design*, Table 14–2.
- P : diametral pitch equivalent [teeth/in]

Load Distribution Factor K_H

The load distribution factor K_H (also denoted K_m) accounts for the nonuniform distribution of load across the gear face width caused by shaft deflection, bearing spacing, manufacturing inaccuracies, and mounting conditions. Uniform load distribution along the face width is rarely achieved in practice, particularly in compact, high-torque gearboxes.

Under AGMA assumptions (gear elements mounted between bearings and contact occurring across the full face width of the narrowest member), the load distribution factor is expressed as:

$$K_H = K_m = 1 + C_{mc} (C_{pf} C_{pm} + C_{ma} C_e) \quad (22)$$

- **Tooth crowning factor**

$$C_{mc} = \begin{cases} 1.0 & \text{for uncrowned teeth} \\ 0.8 & \text{for crowned teeth} \end{cases} \quad (23)$$

- **Face width factor**

$$C_{pf} = \begin{cases} \frac{F}{10d_p} - 0.025, & F \leq 1 \text{ in} \\ \frac{F}{10d_p} - 0.0375 + 0.0125F, & 1 < F \leq 17 \text{ in} \\ \frac{F}{10d_p} - 0.1109 + 0.0207F - 0.000228F^2, & 17 < F \leq 40 \text{ in} \end{cases} \quad (24)$$

where F is the face width (in) and d_p is the pinion pitch diameter (in). For values of $F/(10d_p) < 0.05$, a minimum value of 0.05 is used.

- **Mounting factor**

$$C_{pm} = \begin{cases} 1.0 & \text{for straddle-mounted pinion with } S_1/S < 0.175 \\ 1.1 & \text{for straddle-mounted pinion with } S_1/S \geq 0.175 \end{cases} \quad (25)$$

where S is the total distance between 2 bearings and S_1 is the distance from gear face to the center.

- **Manufacturing and alignment factor**

$$C_{ma} = A + BF + CF^2 \quad (26)$$

where the coefficients A , B , and C depend on the gear manufacturing quality and are selected from AGMA reference tables, provided in *Shigley's Mechanical Engineering Design*, Table 14–9.

- **Assembly factor**

$$C_e = \begin{cases} 0.8 & \text{for gears adjusted at assembly or lapped} \\ 1.0 & \text{for all other conditions} \end{cases} \quad (27)$$

The factor K_H is applied in both bending and contact stress equations and plays a critical role in accounting for load misalignment effects in real gearbox assemblies.

Rim Thickness Factor K_B

Assumed I am dealing with constant thickness gears, the resulting rim thickness factor $K_B = 1$.

Geometry Factor for Bending Strength Y_J

This factor depends on the gear's pressure angle, the module and the number of teeth on both the pinion and the mating gear.

These values can be obtained with the help of Figure 14–6, in *Shigley's Mechanical Engineering Design*.

5.3 Safety Factors Results

Table 11: Summary of Gears Safety Factors

Gear	Module	Number of teeth	Face width [mm]	Bending safety factor	Contact safety factor (failure by wear)
Gear 1 input	3.0	25	40	1.2832	1.3155
Gear 1 output	3.0	63	40	1.3008	1.3419
Gear 2 input	3.0	31	30	1.2621	1.3273
Gear 2 output	3.0	57	30	1.2710	1.3434
Gear 3 input	3.0	37	25	1.2966	1.3658
Gear 3 output	3.0	51	25	1.3015	1.3746
Gear 4 input	3.0	41	25	1.4347	1.4366
Gear 4 output	3.0	47	25	1.4370	1.4405
Gear 5 input	3.0	49	20	1.3997	1.3961
Gear 5 output	3.0	39	20	1.3960	1.3897
Reverse gear in	2.5	31	40	1.2627	1.4319
Reverse gear out	2.5	68	40	1.2767	1.4558

6 Keys

6.1 General Description

Torque transmission between the shafts and the mounted gears in the sequential gearbox is ensured using non-standard parallel keys. Unlike conventional catalog keys, these keys were specifically dimensioned and verified by calculation to match the transmitted torque levels and packaging constraints of the gearbox.

All keys used in this design are manufactured from **AISI 4340 steel in the normalized condition**, with the following mechanical properties:

- Ultimate tensile strength: $S_{ut} = 1110$ MPa
- Yield strength: $S_y = 710$ MPa

Failure at keys can be 2 types: failure by crushing and failure by shear. Both safety factors should meet the requirements to consider the key as acceptable. More details in the upcoming paragraphs.

6.2 Torque Transmission

The torque transmitted through the shaft produces a tangential force acting on the key, given by:

$$F_t = \frac{2T}{d} \quad (28)$$

where:

- T is the transmitted torque,
- d is the shaft diameter.

This tangential force is responsible for both shear stress within the key and bearing stress at the key–shaft and key–hub interfaces.

6.3 Shear Failure of the Key

The average shear stress in the key is expressed as:

$$\tau = \frac{F_t}{bl} \quad (29)$$

where:

- b is the key width,
- l is the effective key length.

The allowable shear stress for ductile steel is taken as:

$$\tau_{\text{allow}} = 0.30 \sigma_y \quad (30)$$

The shear safety factor is therefore:

$$n_{\text{shear}} = \frac{\tau_{\text{allow}}}{\tau} \quad (31)$$

6.4 Crushing (Bearing) Failure of the Key

The bearing stress acting on the key is given by:

$$\sigma_c = \frac{F_t}{hl} \quad (32)$$

where h is the effective bearing height of the key. Both the key–shaft and key–hub interfaces are considered in the evaluation.

The allowable bearing stress is defined as:

$$\sigma_{c,\text{allow}} = 0.6 S_y \quad (33)$$

The corresponding crushing safety factor is:

$$n_{\text{crushing}} = \frac{\sigma_{c,\text{allow}}}{\sigma_c} \quad (34)$$

6.5 Conclusion

All keys in the gearbox were verified against shear and crushing failure using the above criteria. The use of high-strength AISI 4340 normalized steel ensures sufficient load-carrying capacity, desired safety factor levels and packaging requirements of the sequential gearbox design.

Results are tabulated below (an extended table can be found in the additional documents).

Table 12: Safety Factors of Keyed Connections

Location	n_{shear}	$n_{\text{crushing,shaft}}$	$n_{\text{crushing,hub}}$	Governing Mode
Input shaft – reverse gear	2.84	2.92	3.15	Shear
Input shaft – gear 1	3.55	3.65	3.94	Shear
Input shaft – gear 2	2.60	2.68	2.89	Shear
Input shaft – gear 3	2.60	2.68	2.89	Shear
Input shaft – gear 4	2.37	2.43	2.62	Shear
Input shaft – flange	1.78	1.82	1.80	Shear
Dog ring sleeve (2–3)	1.95	1.67	1.72	Crushing
Dog ring sleeve (4–5)	2.58	2.21	2.28	Crushing

The target safety factor is around $n = 1.3 - 1.5$ as the design target for all keyed connections. All evaluated key locations satisfy this criterion with comfortable margins.

Increasing the safety factor of keyed joints does not introduce any penalty in terms of mass, packaging, or dynamic behavior. On the contrary, keys are simple, low-cost, and easily replaceable components, making conservative design choices acceptable and even desirable.

In certain locations, the yield strength used in the crushing verification may correspond to either the key material, the shaft material, or the hub material. This choice can slightly affect the numerical value of the crushing safety factor. However, regardless of this assumption, all results remain well within the acceptable design range, and no keyed joint approaches a critical failure condition.

7 Splines

7.1 General Description

Spline connections are used in the sequential gearbox to transmit torque while allowing axial sliding between components, notably for the dog ring sleeves. Unlike keys, splines distribute the transmitted torque over multiple teeth, leading to improved load sharing and reduced stress concentration.

All spline connections in this gearbox were selected and dimensioned in accordance with the **DIN 5480 standard**, which defines the geometry, tolerances, and load-carrying capacity of involute splines. The spline sizes were chosen based on the transmitted torque, shaft diameter, and packaging constraints, ensuring compatibility with standardized manufacturing practices and reliable load distribution.

7.2 Load Transmission Mechanism

In a splined joint, torque is transmitted through compressive contact(bearing) between the spline teeth of the shaft and the mating hub. The load is distributed over several teeth simultaneously, depending on manufacturing tolerances, elastic deformation, and contact conditions.

The tangential force corresponding to the transmitted torque is given by:

$$F_t = \frac{2T}{d_m} \quad (35)$$

where:

- T is the transmitted torque,
- d_m is the mean spline diameter.

This force is shared among the engaged spline teeth.

7.3 Failure Mode Consideration

For spline connections, only crushing (bearing) failure is considered in the design verification. Shear failure is not evaluated because:

- Torque transmission does not rely on a single shear plane, as in keyed joints
- The spline teeth are integral to the shaft and hub, eliminating a discrete shear element
- Load transfer occurs primarily through surface contact pressure rather than material shearing.

Consequently, the governing failure mechanism for splines is crushing(bearing) stress at the tooth contact surfaces.

7.4 Crushing (Bearing) Stress in Splines

The average bearing stress in the splines is expressed as:

$$\sigma_c = \frac{F_t}{N A_c} \quad (36)$$

where:

- N is the number of load-carrying spline teeth,
- A_c is the effective contact area per tooth.

The allowable bearing stress is taken as a fraction of the yield strength of the weaker material between the shaft and the hub:

$$\sigma_{c,allow} = 0.6 S_y \quad (37)$$

The crushing safety factor is therefore:

$$n_{crushing} = \frac{\sigma_{c,allow}}{\sigma_c} \quad (38)$$

7.5 Conclusion

All spline connections were verified against crushing failure and satisfy the design requirement with safety factors greater than the target value. Results are tabulated below.

Table 13: Crushing Safety Factors of Spline Connections

Location	$n_{crushing}$
Input shaft – gear 5	6.64
Output shaft – gear 1 / reverse	1.60
Dog ring sleeve (gears 2–3)	3.84
Dog ring sleeve (gears 4–5)	3.84
Output shaft flange	3.22

8 Bearings

8.1 Bearing Life Theory

Bearing life is evaluated using the basic rating life formulation as defined in Shigley's *Mechanical Engineering Design*. The basic rating life, expressed in millions of revolutions, is given by:

$$L_{10} = \left(\frac{C}{P} \right)^p \quad (39)$$

where:

- L_{10} is the basic rating life (in 10^6 revolutions)
- C is the basic dynamic load rating of the bearing
- P is the equivalent dynamic bearing load
- p is the life exponent ($p = 3$ for ball bearings, $p = \frac{10}{3}$ for roller bearings)

8.2 Equivalent Dynamic Load

For combined radial and axial loading, the equivalent dynamic bearing load is defined as:

$$P = XVF_r + YF_a \quad (40)$$

where:

- F_r is the radial bearing load
- F_a is the axial bearing load
- X and Y are load factors dependent on the bearing type and load ratio (with $X = 1$ and $Y = 0$ for spur gears)

In the expression of the equivalent dynamic bearing load, a rotation factor V is introduced to account for the rotation of the inner or outer ring. For most rolling-element bearings operating under normal conditions, this factor is taken as $V = 1$. The value $V = 1.2$ is occasionally used to account for unfavorable outer-ring rotation, as it reflects a reduction in fatigue life under such conditions.

In the present gearbox design, main bearings operate with a rotating inner ring and a stationary outer ring. Therefore, the rotation factor is taken as $V = 1$, in accordance with standard practice and the recommendations provided in Shigley's *Mechanical Engineering Design*.

8.3 Bearing Life in Operating Hours

The bearing life expressed in operating hours is obtained from:

$$L_{10h} = \frac{10^6 L_{10}}{60n} \quad (41)$$

where:

- L_{10h} is the bearing life in hours
- n is the rotational speed in revolutions per minute (I chose 4000 rpm as the average gearbox's shaft rotation speed)

8.4 Variable Load and Speed Conditions

When bearings operate under variable loads and speeds, an equivalent bearing load is defined using a time-weighted formulation:

$$P_{eq} = \left(\frac{\sum_i n_i P_i^p t_i}{\sum_i n_i t_i} \right)^{1/p} \quad (42)$$

where:

- P_i is the equivalent dynamic load during operating condition i
- t_i is the time spent in condition i
- p is the bearing life exponent

This equivalent load is then used in the basic rating life equation.

8.5 Design Criterion

For each bearing location, an equivalent dynamic load P_{eq} is first computed from the duty cycle. The bearing is then selected from manufacturer catalogs such that its basic dynamic load rating C_{10} satisfies:

$$C_{10} \geq P_{eq} \quad (43)$$

This criterion ensures that the chosen bearing has sufficient dynamic load capacity for the intended operating conditions. A bearing with a higher C_{10} than P_{eq} provides additional life margin.

8.6 Results and Bearing Selection

After applying the duty-cycle-based formulation, the equivalent dynamic bearing loads were found to be:

- Input shaft: $P_{eq} = 43.8$ kN
- Output shaft: $P_{eq} = 38.2$ kN

After consulting manufacturer catalogs, the following cylindrical roller bearings were selected for use in the gearbox:

- NUP 2206 ECP (bore diameter 30 mm, width 20 mm, $C_{10} = 55$ kN)
- NUP 207 ECP (bore diameter 35 mm, width 17 mm, $C_{10} = 56$ kN)

- NUP 208 ECP (bore diameter 40 mm, width 18 mm, $C_{10} = 62$ kN)

All selected bearings have basic dynamic load ratings exceeding the calculated equivalent loads, thereby satisfying the adopted design criterion ($C_{10} > P_{eq}$). The use of cylindrical roller bearings is well suited to the high radial loads encountered in the gearbox and provides adequate margin with respect to bearing life.

8.7 Other Bearings

Other bearings were also used in this gearbox design. These bearings are located between the output shaft and output gears bore and they are used to allow free rotation of output gears, All those bearings were chosen from catalogs without additional calculations. I mainly chose NTN needle bearings, and below, a table that contain each gear with its corresponding bearing.

Table 14: Needle Bearings Used for Gear Support

Gear location	Needle bearing reference
Output gear 1	NA6190R
Output gear 2	NA5909
Output gear 3	NK50_25RCT + IR45x50x25
Output gear 4	NA4908R
Output gear 5	NK45_20R + IR40x45x20
Reverse gear	NA6190R

9 Engagement Mechanism

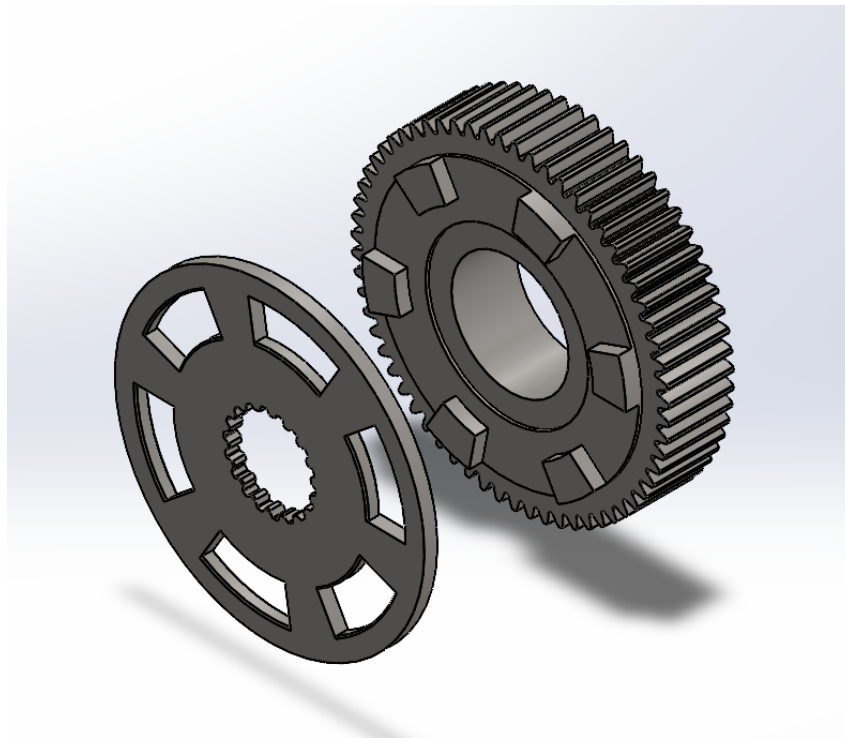


Figure 2: Dog ring engagement with gear dog teeth

9.1 Principle of Operation

Gear engagement in the sequential gearbox is achieved using a **dog ring mechanism**. Each selectable gear is permanently in mesh with its mating gear, while torque transmission is controlled by axial engagement of a sliding dog ring.

The dog ring is splined to the shaft, allowing it to slide axially while remaining rotationally constrained. When actuated, the dog ring moves along the shaft and engages with the dog teeth machined on the face of the gear. Once engaged, torque is transmitted directly through the contacting dog teeth, locking the gear to the shaft.

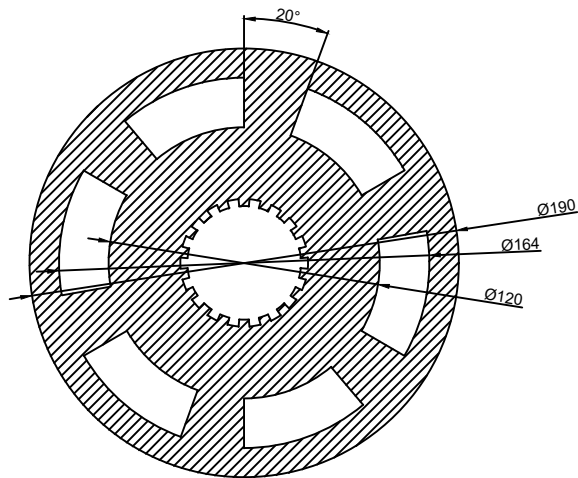
9.2 Engagement and Disengagement

During a gear change, the dog ring disengages from the currently engaged gear and slides axially toward the target gear. Engagement occurs when the dog teeth of the ring align and enter the corresponding recesses on the gear face. Successful engagement results in a rigid mechanical connection between the gear and the shaft. Disengagement is achieved by reversing the axial motion of the dog ring, freeing the gear while it remains in constant mesh with its counterpart.

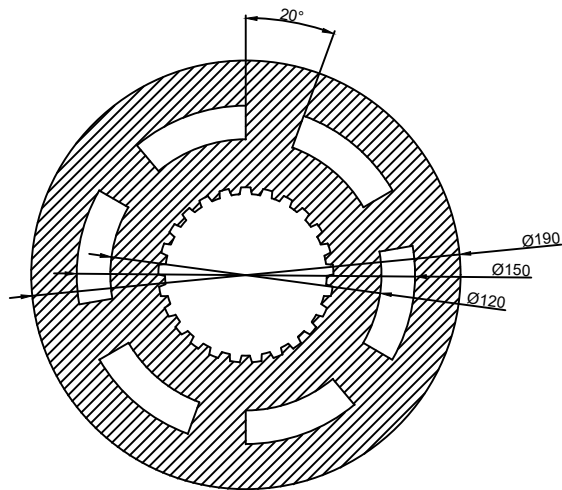
9.3 Design Characteristics

Unlike synchronizers, the dog ring mechanism does not rely on friction to match rotational speeds. Engagement therefore occurs under conditions of reduced load or momentary torque interruption, which is consistent with the operating principle of a sequential gearbox. The simplicity and robustness of the dog ring mechanism make it well suited for performance-oriented applications, where rapid gear changes and positive engagement are required.

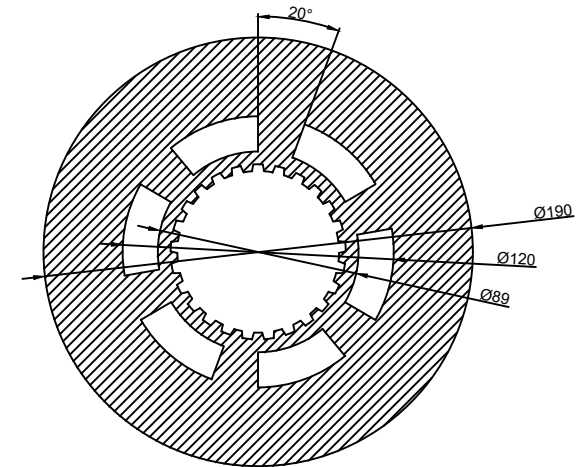
Below is a simple sketch of all dog rings designed and used in the gearbox.



Dog Ring Gear Rev - 1
 Reverse gear root diameter = 163.5
 module = 3
 spline pitch diameter = 54
 root diameter = 50.1
 tip diameter = 56.7
 Dog pocket outer diameter = 164
 Dog pocket inner diameter = 120
 Dog angle = 20



Dog Ring Gear 2 - 3
 Gear 3 root diameter = 145.5
 module = 3
 spline pitch diameter = 75
 root diameter = 71.7
 tip diameter = 77.7
 Dog pocket outer diameter = 150
 Dog pocket inner diameter = 120
 Dog angle = 20



Dog Ring Gear 4 - 5
 Gear 5 root diameter = 109.5
 module = 3
 spline pitch diameter = 75
 root diameter = 71.7
 tip diameter = 77.7
 Dog pocket outer diameter = 110
 Dog pocket inner diameter = 89
 Dog angle = 20

9.4 Dog Clutch Failure Checks

Dog clutches are checked against crushing failure of the dog faces and shear failure of the dogs. Torque transmission is assumed to occur through face contact between engaged dogs.

9.4.1 Tangential Force at Dog Radius

The transmitted torque T produces a tangential force at the mean dog radius:

$$F_t = \frac{T}{r_m} \quad (44)$$

where r_m is the mean radius of the dog engagement.

9.4.2 Crushing Failure of Dog Faces

The average bearing (crushing) stress on the engaged dog faces is:

$$\sigma_c = \frac{F_t}{N_e A_d} \quad (45)$$

where:

- A_d is the contact area of one dog
- N_e is the effective number of dogs carrying load (value is usually somewhere between 0.25-0.3, for this gearbox design I chose 0.3)

The crushing safety factor is defined as:

$$n_{\text{crushing}} = \frac{\sigma_{c,\text{allow}}}{\sigma_c} \quad (46)$$

with:

$$\sigma_{c,\text{allow}} = 0.6 S_y \quad (47)$$

9.4.3 Shear Failure of Dogs

Each dog is also checked in shear at its root section. The average shear stress in one dog is:

$$\tau = \frac{F_t}{N_e A_s} \quad (48)$$

where A_s is the shear area of one dog (root cross-section).

The shear safety factor is given by:

$$n_{\text{shear}} = \frac{\tau_{\text{allow}}}{\tau} \quad (49)$$

with:

$$\tau_{\text{allow}} = 0.3 S_y \quad (50)$$

9.4.4 Results

Safety factors obtained are tabulated below.

Table 15: Dog Ring Geometry and Safety Factors

Dog ring	Number of dogs/teeth	Axial length	Radial width	Arc length	Failure by shear	Failure by crushing
Reverse – gear 1	6	8	20	7.9744	4.2835	8.5946
Gear 2 – 3	6	8	10	8.4569	2.9354	5.5535
Gear 4 – 5	6	8	10	11.5167	4.7984	6.6663

10 Design Summary

The sequential gearbox was designed following a structured and coherent approach. Gear geometry was defined to satisfy torque and speed requirements, while shafts were dimensioned and verified against combined loading conditions. Bearing selection was performed using duty-cycle-based equivalent loads, and all selected bearings satisfy the adopted dynamic load rating criterion.

Torque transmission elements, including keys, splines, and dog clutches, were verified against their governing failure modes with safety factors exceeding the design targets. The engagement mechanism ensures positive gear selection while maintaining mechanical simplicity and robustness.

11 Design Assumptions and Limitations

The present design is based on analytical calculations and standardized design methods. Dynamic effects during gear engagement, manufacturing tolerances, and thermal effects were not explicitly modeled. Load sharing assumptions were made for splines and dog clutches, and fatigue effects were not considered for engagement components.

Despite these simplifications, the design remains representative of a realistic sequential gearbox and provides sufficient safety margins for the intended operating conditions.

12 Supporting Material

For clarity and conciseness, the main body of this report presents only the most critical design cases. Additional tables containing detailed calculations, secondary loading cases, and extended design data are provided alongside this report.

Furthermore, complete technical drawings of the components, as well as assembly models and views of the gearbox, are included as supporting material. These documents ensure full traceability of the design and provide the necessary information for further analysis.

13 Conclusion

This project was carried out with the objective of developing a sequential gearbox design that could realistically be taken beyond the conceptual stage and into manufacturing. All verification and validation performed within this report are based on analytical and theoretical methods; no experimental testing or physical validation has been conducted at this stage. As such, the present work should be regarded as a solid design foundation rather than a finalized, validated product.

Beyond the technical results, this project served as an in-depth learning experience in mechanical design. Particular emphasis was placed on understanding design assumptions: when they are required, when they are valid, and how they affect the interpretation of results. The project required continuous judgment in balancing analytical rigor with practical simplifications, reflecting the reality of real-world engineering design.

A wide range of drivetrain mechanisms had to be thoroughly understood in order to be correctly implemented, including gears, shafts, bearings, splines, keys, dog clutches, and engagement mechanisms. Unlike conventional reducers or speed-increasing gearboxes with fixed operating conditions, this sequential gearbox combines multiple operating cases within a single architecture. Each gear engagement represents a distinct load case, requiring careful evaluation of the most critical conditions while ensuring overall design coherence.

Countless hours were invested in iterating the design, studying standards, verifying critical cases, and refining the gearbox architecture. This work represents a deliberate effort to bridge theoretical knowledge with practical mechanical design, and it provides a strong basis for further refinement, manufacturing, or experimental validation.